THE FRIDAY UNIVERSE

DEFINITIVE MASTER ARCHITECTURE & TECHNICAL CODEX — SPECIFICATION V3.0.0

Authoritative Engineering Specification: Unified 10-Agent Sovereign AI Operating System, Dual AMD Micro Server Fleet (140.245.208.100 & 140.245.243.168), Dual Oracle Cloud Autonomous 23ai Relational & Vector Stores, Zero-Disk In-Memory Persistence, and Granular GitHub Actions CI/CD Synchronization.

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1. Executive Summary & Sovereign Multi-Agent Philosophy

The FRIDAY Universe is an enterprise-grade autonomous cognitive architecture comprising ten specialized agent subsystems. The system operates across high-efficiency Oracle Cloud Infrastructure (OCI) Always Free hardware ($0.00 compute boundary) with zero persistent SQLite or PostgreSQL files on local VM disks. All relational state, vector embeddings, and audit trails reside in two Oracle Autonomous 23ai cloud databases.

FIVE NON-NEGOTIABLE OPERATIONAL INVARIANTS

1. **Strict Proper Naming Only:** Exact names: FRIDAY, Inference, Memora, Stratex, Futuris, IntelX, Forge, VyralX, Sentinel, Cortex. No hyphens or abbreviations.
2. **Zero Local Disk SQLite/Postgres:** No .db or .sqlite files on persistent VM storage. Temporary in-memory buffers strictly use Linux tmpfs (/dev/shm in RAM).
3. **1 GB Physical RAM Budget:** Both AMD micro VMs are capped at 1 GB RAM each. Single-worker Uvicorn (--workers 1), explicit cgroup limits (MemoryMax), and zero memory bloat.
4. **Systemd Specifier Escaping:** Database passwords with %21%23 must be stored in /etc/default/friday_universe.env. Never place them inline in .service files.
5. **Fail-Closed Security & Immutability:** Forge code modifications require Sentinel verification before merge. Return new objects rather than mutating state.

2. Dual-Server Infrastructure Topography

The 10 autonomous agents are physically distributed across two Oracle Cloud Always Free AMD Micro VMs (VM.Standard.E2.1.Micro, 1 OCPU, 1 GB RAM, 8 GB swapfile) in the AP-Hyderabad-1 region:

Fleet Tier	Public IP / Hostname	Private IP	Hosted Agents & Mesh Ports	Memory Allocation & Cgroup Caps
Server 1 **Core Fleet** (The Brain)	140.245.208.100 friday-amd-vnic	10.0.0.121	• **FRIDAY** (:10000) — Core Gateway • **Inference** (:8001) — Model Router • **Memora** (:8002) — Vector Memory • **Stratex** (:8003) — Trading Desk • **Futuris** (:8004) — Forecasting • **IntelX** (:8005) — Deep Research • **Forge** (:8006) — Code Synthesizer	**Physical RAM:** 956 MB (1 GB) **Swapfile:** 8,191 MB (8 GB) **FRIDAY:** MemoryMax=200M **Memora:** MemoryMax=200M **Others:** MemoryMax=180M each **Target DB:** friday_universe_db
Server 2 **Specialist Fleet** (Studio & Ops)	140.245.243.168 friday-amd-2-vnic	10.0.0.120	• **VyralX** (:8010) — Video Production • **Sentinel** (:8008) — Security Shield • **Cortex** (:8009) — CRM & Web Engine	**Physical RAM:** 956 MB (1 GB) **Swapfile:** 8,191 MB (8 GB) **VyralX:** MemoryMax=250M **Sentinel:** MemoryMax=200M **Cortex:** MemoryMax=200M **Target DB:** vyralx_db (VyralX) / Core (Others)

Systemd Service Execution Table

Agent	Server	Port	Systemd Service	Working Directory & Execution Command	Memory Limit
FRIDAY	Server 1	10000	FRIDAY.service	/opt/friday_universe/FRIDAY /opt/friday_universe/venv/bin/python -m uvicorn friday.api.server:app --host 0.0.0.0 --port 10000 --workers 1	200M
Inference	Server 1	8001	Inference.service	/opt/friday_universe/Inference /opt/friday_universe/venv/bin/python -m uvicorn app.main:app --host 0.0.0.0 --port 8001 --workers 1	180M
Memora	Server 1	8002	Memora.service	/opt/friday_universe/Memora /opt/friday_universe/venv/bin/python -m uvicorn app.main:app --host 0.0.0.0 --port 8002 --workers 1	200M
Stratex	Server 1	8003	Stratex.service	/opt/friday_universe/Stratex /opt/friday_universe/venv/bin/python -m uvicorn main:app --host 0.0.0.0 --port 8003 --workers 1	180M
Futuris	Server 1	8004	Futuris.service	/opt/friday_universe/Futuris /opt/friday_universe/venv/bin/python -m uvicorn futuris.api.app:app --host 0.0.0.0 --port 8004 --workers 1	180M
IntelX	Server 1	8005	IntelX.service	/opt/friday_universe/IntelX /opt/friday_universe/venv/bin/python -m uvicorn intelx.app.main:app --host 0.0.0.0 --port 8005 --workers 1	180M
Forge	Server 1	8006	Forge.service	/opt/friday_universe/Forge /opt/friday_universe/venv/bin/python -m uvicorn app.main:app --host 0.0.0.0 --port 8006 --workers 1	180M
VyralX	Server 2	8010	VyralX.service	/opt/friday_universe/VyralX /opt/friday_universe/venv/bin/python -m uvicorn app.main:app --host 0.0.0.0 --port 8010 --workers 1	250M
Sentinel	Server 2	8008	Sentinel.service	/opt/friday_universe/Sentinel /opt/friday_universe/venv/bin/python -m uvicorn sentinel.apps.api.main:app --host 0.0.0.0 --port 8008 --workers 1	200M
Cortex	Server 2	8009	Cortex.service	/opt/friday_universe/Cortex /opt/friday_universe/venv/bin/python -m uvicorn apps.api.src.cortex_api.main:app --host 0.0.0.0 --port 8009 --workers 1	200M

3. Dual Oracle Autonomous 23ai Database Architecture

All persistent state across all agents lives in two Always Free Oracle Autonomous Transaction Processing (ATP) 23ai instances in ap-hyderabad-1 connecting over TLS Port 1521 (one-way TLS; no client wallet required). Zero local database files are written to VM disk.

Database Identifier	Service Name & DSN Details	Primary Owner	Dedicated Tenancy	Core Schema Tables
friday_universe_db (Core Fleet DB)	g7e6177242670c3_fridaycore_medium.adb.oraclecloud.com Host: adb.ap-hyderabad-1.oraclecloud.com Port: 1521 (Protocol: TCPS, SSL match: yes)	ADMIN FridayUniverse2026!#Core	FRIDAY, Inference, Memora, Stratex, Futuris, IntelX, Forge, Sentinel, Cortex	• MEMORA_MEMORIES (23ai HNSW native vector) • FUTURIS_FORECASTS & 12 schema tables • API_KEYS (unified IntelX/Cortex auth) • RESEARCH_RUNS, CLAIMS, EVIDENCE • FORGE_BUILDS, STRATEX_TRADES • SENTINEL_SCANS, AUDIT_EVENTS
vyralx_db (Specialist Media DB)	g7e6177242670c3_fridayvyralx_medium.adb.oraclecloud.com Host: adb.ap-hyderabad-1.oraclecloud.com Port: 1521 (Protocol: TCPS, SSL match: yes)	ADMIN FridayUniverse2026!#Vyral	Exclusively dedicated to **VyralX**	• VYRALX_PROJECTS, VYRALX_SCRIPTS • VYRALX_STORYBOARDS, RENDER_JOBS • VYRALX_MEDIA_ASSETS, PUBLISH_QUEUE • VIDEO_METRICS, AGENT_TASKS

Key Oracle 23ai Architectural Invariants

• **Native 23ai Vector Columns:** Memora utilizes Oracle 23ai native VECTOR data types with HNSW (Hierarchical Navigable Small World) index structures, executing sub-second semantic cosine similarity queries directly inside the database engine.
• **Unified API_KEYS Schema:** IntelX and Cortex share a single consolidated API_KEYS table in friday_universe_db with nullable columns and sensible defaults (tenant_id, site_id, key_prefix, role, is_active, revoked), preventing DDL conflicts.
• **Zero-Disk Linux tmpfs Caching:** Any local SQLite caches (e.g., Inference rate limits, Forge Memora fallback) are stored in /dev/shm (pure RAM tmpfs). No SQLite files touch persistent storage.

4. Deep Technical Subsystem Specifications

Detailed functional roles, technology stacks, internal endpoints, and safety gates for each of the 10 sovereign agents:

FRIDAY — Cognitive Orchestrator & Action Gateway (Port 10000 | Server 1)

Framework Stack: FastAPI, Uvicorn, WebSockets, Asyncio, Pydantic V2
Service Unit: FRIDAY.service
Primary Mission: Central human-in-the-loop orchestrator, task planner, voice/chat companion, and mesh dispatcher.
Key API Routes: POST /api/v1/tasks (task envelope), GET /api/v1/tasks/{id} (status), POST /api/v1/actions/approve (human gate), GET /health

Inference — Sovereign Multi-Model Routing Gateway (Port 8001 | Server 1)

Framework Stack: FastAPI, HTTPX Pool, Groq SDK, Cerebras, OpenRouter, Google GenAI
Service Unit: Inference.service
Primary Mission: High-throughput multi-model inference router with automated fallback across Groq, Cerebras, OpenRouter, and Gemini.
Key API Routes: POST /v1/chat/completions (OpenAI compatible), POST /v1/task/execute (structured reasoning), GET /health

Memora — Cognitive Memory Fabric & Vector Database (Port 8002 | Server 1)

Framework Stack: FastAPI, Oracle Database 23ai Native VECTOR, HNSW Indexing
Service Unit: Memora.service
Primary Mission: Sole sovereign long-term memory authority enforcing cryptographic provenance, multi-tier scoping (Private, Shared, Universal), and prompt injection isolation.
Key API Routes: POST /v1/memories (ingest), POST /v1/memories/search (semantic vector search), GET /health

Stratex — Quantitative Trading Desk & Risk Ledger (Port 8003 | Server 1)

Framework Stack: FastAPI, NumPy, Pandas, CCXT, Binance Futures Testnet SDK
Service Unit: Stratex.service
Primary Mission: Algorithmic execution desk, continuous market surveillance, risk governance, and quantitative backtesting. Zero real-money risk.
Key API Routes: POST /api/v1/orders (pre-trade risk validate), POST /api/v1/emergency-halt (atomic kill-switch), GET /health

Futuris — Predictive Forecasting & Calibration Engine (Port 8004 | Server 1)

Framework Stack: FastAPI, APScheduler, SciPy, Statsmodels, SQLAlchemy 2.0 Async
Service Unit: Futuris.service
Primary Mission: Scenario forecasting, probabilistic modeling, uncertainty estimation, and historical Brier score calibration.
Key API Routes: POST /v1/forecast/scenario (multi-branch forecast), GET /v1/forecast/regime (regime distributions), GET /health (12 tables verified)

IntelX — Deep Evidence Research & Fact Verification (Port 8005 | Server 1)

Framework Stack: FastAPI, BeautifulSoup4, Trafilatura, DuckDuckGo / SearXNG Connectors
Service Unit: IntelX.service
Primary Mission: Deep web research, fact-checking, citation extraction, news ingestion, and claim corroboration against primary sources.
Key API Routes: POST /v1/research/query (deep gather), POST /v1/verify/claim (corroborate assertion), GET /health

Forge — Autonomous Software Synthesizer & Build Engine (Port 8006 | Server 1)

Framework Stack: FastAPI, Rope, Pytest, AST, Tree-Sitter, SQLAlchemy 2.0 Async
Service Unit: Forge.service
Primary Mission: Full-stack autonomous software engineer: specification parsing, multi-file code synthesis, automated test generation, and verification.
Key API Routes: POST /api/v1/synthesize (DAG build), GET /api/v1/tasks/{id}/verify (10-point test battery), GET /health

VyralX — Autonomous Video Production & Multi-Channel Studio (Port 8010 | Server 2)

Framework Stack: FastAPI, MoviePy, FFmpeg, FFprobe, eSpeak NG, YouTube Data API v3
Service Unit: VyralX.service
Primary Mission: End-to-end video production, AI scriptwriting, audio synthesis, visual assembly, automated captioning, rendering, and YouTube publication.
Key API Routes: POST /api/v1/projects (scene plan), POST /api/v1/render (FFmpeg compose), POST /api/v1/publish/youtube (chunked upload), GET /health

Sentinel — Zero-Trust Cybersecurity Gatekeeper & Audit Shield (Port 8008 | Server 2)

Framework Stack: FastAPI, Cryptography, Jose JWT, SSL/TLS Audit Daemons
Service Unit: Sentinel.service
Primary Mission: Ecosystem security officer: cryptographic HMAC audit ledger, continuous TLS/DNS monitoring, static code vulnerability analysis, and zero-trust perimeter defense.
Key API Routes: POST /v1/security/audit (fail-closed code review gate), POST /v1/audit/record (signed audit ledger), GET /health

Cortex — Autonomous Operations Engine & Governed CRM (Port 8009 | Server 2)

Framework Stack: FastAPI, Next.js (Exported SPA in apps/dashboard/out), SQLAlchemy Async
Service Unit: Cortex.service
Primary Mission: Powers ecosystem web presence, landing pages, operator dashboard API, CRM telemetry, and event-driven self-healing.
Key API Routes: GET / (landing page & status), POST /v1/events (telemetry ingest), GET /health (tenant isolation verify)

5. Unified GitHub Monorepo & Granular CI/CD Deployment

Development is centralized in a unified monorepo while deployment and standalone repository distribution are handled through automated GitHub Actions workflows.

- **Monorepo Remote:** https://github.com/surendra2304/FRIDAY-Universe.git (Branch: main)
- **Local Workspace Path:** D:\FRIDAY Universe
- **Remote Target Path:** /opt/friday_universe on both Server 1 and Server 2.

Workflow File	Trigger Condition	Primary Mechanism	Deployment Target & Action
deploy-monorepo.yml (Selective Fleet Deploy)	Push to main (Path-filtered)	dorny/paths-filter@v3 detects modified agent subdirectories (e.g. FRIDAY/**, VyralX/**). Executes parallel SSH tasks via appleboy/ssh-action.	Connects to Server 1 or Server 2 based on agent location. Pulls updated code into /opt/friday_universe/ and triggers sudo systemctl restart .service. Unchanged agents remain running without restart.
mirror-subrepos.yml (Subtree Synchronization)	Push to main (All commits)	Executes git subtree split across all 10 agent folders and force-pushes the clean commit history directly to each standalone repository.	Syncs to 10 distinct GitHub repositories under surendra2304/.git, maintaining isolated histories for independent packaging.

6. Proven Production Coding Gotchas & Self-Healing Runbook

All engineers and AI agents modifying FRIDAY Universe code must adhere to these battle-tested resolutions to avoid production crashes:

Issue / Error Signature	Root Technical Cause	Enforced Architecture Resolution
SQLAlchemy Oracle JSON Compile Error Compiler can't render element of type JSON	SQLAlchemy 2.0 OracleTypeCompiler lacks a native handler for generic JSON / JSONB types.	Register the native Oracle 23ai compiler hook in model definitions: @compiles(JSON, 'oracle') def _compile(t, c, **kw): return 'JSON'
Oracle JSON Path Parameter Error ORA-40454: path expression not a literal	SQLAlchemy binds JSON dictionary indexing (json_getitem_op) as a bind variable, which Oracle SQL rejects in JSON_VALUE.	Wrap path expressions in literal_column when querying Oracle: func.json_value(col, literal_column('"$.priority"'))
Systemd Credential Specifier Bug Failed to resolve specifiers ... invalid slot	Systemd interprets % in passwords (e.g. %21%23) as format specifiers and discards the line.	Store credentials in /etc/default/friday_universe.env and reference them via EnvironmentFile=/etc/default/.... Never inline.
Async Dialect Rejection create_async_engine requires async driver	Passing oracle+oracledb:// to create_async_engine() fails because async execution requires the async dialect.	Dynamically convert connection strings: if url.startswith('oracle+oracledb://'): url = 'oracle+oracledb_async://' + url[len('oracle+oracledb://'):]
Shared API_KEYS Conflict ORA-00904: API_KEYS.REVOKED invalid identifier	IntelX and Cortex both declare an api_keys table with different column definitions in friday_universe_db.	The table in friday_universe_db is unified with a superset schema. Non-shared columns are nullable and carry sensible defaults.

7. Inter-Agent RPC Contracts & Environment Catalog

Inter-agent communication flows over internal private IP REST endpoints with required mutual authentication headers:

Caller Agent	Target Agent	Target Private URL & Port	Action & Protocol	Security Policy
FRIDAY	Inference	http://10.0.0.121:8001/ask	Task reasoning & prompt decomposition	Bearer token authentication
FRIDAY	Memora	http://10.0.0.121:8002/api/v1/memories/search	Vector semantic similarity search	Bearer token + Scoped ACL
Stratex	Futuris	http://10.0.0.121:8004/v1/forecast/regime	Query market regime probability	Advisory insight only
Stratex	Sentinel	http://10.0.0.120:8008/v1/audit/record	Submit trade audit record for HMAC signing	Tamper-evident cryptographic ledger
Forge	Sentinel	http://10.0.0.120:8008/v1/security/audit	Pre-delivery code vulnerability audit	Strict fail-closed veto power
Cortex	IntelX	http://10.0.0.121:8005/v1/research/query	Competitor & factual web research	Asynchronous poll; cited confidence
VyralX	Inference	http://10.0.0.121:8001/v1/task/execute	Original video script & scene generation	Graceful cached draft fallback

Master Environment Variable Catalog (/etc/default/friday_universe.env)

```
# Database Credentials (Loaded via EnvironmentFile in systemd units)
DATABASE_URL=oracle+oracledb://ADMIN:FridayUniverse2026%21%23Core@(description=(retry_count=20)(retry_delay=3)(address=(protocol=tcps)(
port=1521)(host=adb.ap-hyderabad-1.oraclecloud.com))(connect_data=(service_name=g7e6177242670c3_fridaycore_medium.adb.oraclecloud.com))
(security=(ssl_server_dn_match=yes)))
ORACLE_CORE_URL=oracle+oracledb://ADMIN:FridayUniverse2026%21%23Core@(description=(retry_count=20)(retry_delay=3)(address=(protocol=tcp
s)(port=1521)(host=adb.ap-hyderabad-1.oraclecloud.com))(connect_data=(service_name=g7e6177242670c3_fridaycore_medium.adb.oraclecloud.co
m))(security=(ssl_server_dn_match=yes)))
ORACLE_CORE_ASYNC_URL=oracle+oracledb_async://ADMIN:FridayUniverse2026%21%23Core@(description=(retry_count=20)(retry_delay=3)(address=(
protocol=tcps)(port=1521)(host=adb.ap-hyderabad-1.oraclecloud.com))(connect_data=(service_name=g7e6177242670c3_fridaycore_medium.adb.or
aclecloud.com))(security=(ssl_server_dn_match=yes)))
ORACLE_VYRAL_URL=oracle+oracledb://ADMIN:FridayUniverse2026%21%23Vyral@(description=(retry_count=20)(retry_delay=3)(address=(protocol=t
cps)(port=1521)(host=adb.ap-hyderabad-1.oraclecloud.com))(connect_data=(service_name=g7e6177242670c3_fridayvyralx_medium.adb.oracleclou
d.com))(security=(ssl_server_dn_match=yes)))
ORACLE_VYRAL_ASYNC_URL=oracle+oracledb_async://ADMIN:FridayUniverse2026%21%23Vyral@(description=(retry_count=20)(retry_delay=3)(address
=(protocol=tcps)(port=1521)(host=adb.ap-hyderabad-1.oraclecloud.com))(connect_data=(service_name=g7e6177242670c3_fridayvyralx_medium.ad
b.oraclecloud.com))(security=(ssl_server_dn_match=yes)))

# Cryptographic Security Keys
SENTINEL_AUDIT_HMAC_KEY=c36209e9fbe54d008d51a660d2ecb0b023f9906649f8269e8bb95b217dc3ad3e
SENTINEL_AUDIT_SIGNING_KEY=c36209e9fbe54d008d51a660d2ecb0b023f9906649f8269e8bb95b217dc3ad3e
SENTINEL_API_KEY=sentinel_jF-bgAo6XWDQqV-TaRW1MiG36QbC3YUY6c

# Private Mesh Network URLs (Internal IP Interconnect)
FRIDAY_URL=http://10.0.0.121:10000
INFERENCE_URL=http://10.0.0.121:8001
MEMORA_URL=http://10.0.0.121:8002
STRATEX_URL=http://10.0.0.121:8003
FUTURIS_URL=http://10.0.0.121:8004
INTELX_URL=http://10.0.0.121:8005
FORGE_URL=http://10.0.0.121:8006
VYRALX_URL=http://10.0.0.120:8010
SENTINEL_URL=http://10.0.0.120:8008
CORTEX_URL=http://10.0.0.120:8009
```

Document certified and compiled by Antigravity AI Project Coordinator for the FRIDAY Universe. All specifications, schemas, endpoints, and architectural invariants documented herein represent the authoritative ground truth for sovereign ecosystem continuity.